Elliptic curves and isogenies in characteristic 2

Gustavo Banegas, *Nicolas Sarkis*, Benjamin Smith

Inria Saclay, GRACE team

April 1st, 2026 — Journées C2, Côte d'Opale

Motivation

- General context for elliptic curves E/k: $\operatorname{char} k \neq 2, 3$.
- However, working over $k = \mathbb{F}_{2^m}$ is natural computer-wise.

✓ Efficient arithmetic

✗ Only Vélu formulas in char. 2

Figure: RFID tag using ECC

$$\begin{tikzcd} E \arrow[r, "f_A"] \arrow[d, "f_B"'] & E_A \arrow[d, "f_{AB}"] \\ E_B \arrow[r, "f_{BA}"'] & E_{AB} \end{tikzcd}$$

Figure: SIDH protocol (attacked in 2022)

Outline

1. Elliptic curves
2. Isogenies
   - Definition
   - Formulas
3. Conclusion

Elliptic curves

Elliptic curves

Assume char $k \neq 2, 3$:

- Short Weierstrass:

$$E : y^2 = x^3 + ax + b$$

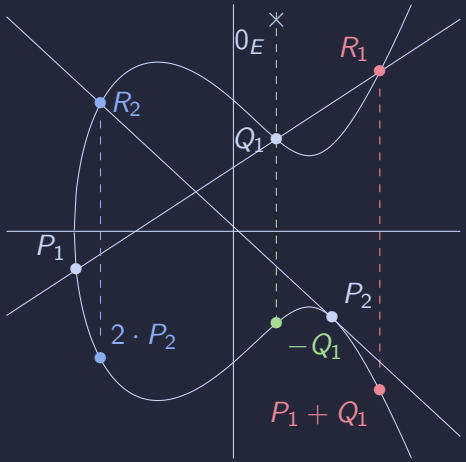

Figure: An elliptic curve

Elliptic curves

Assume char $k = 2$:

- Short Weierstrass:

$$E : \cancel{y^2 = x^3 + ax + b}$$

- Singular in characteristic 2!
- Work with long Weierstrass equations:

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

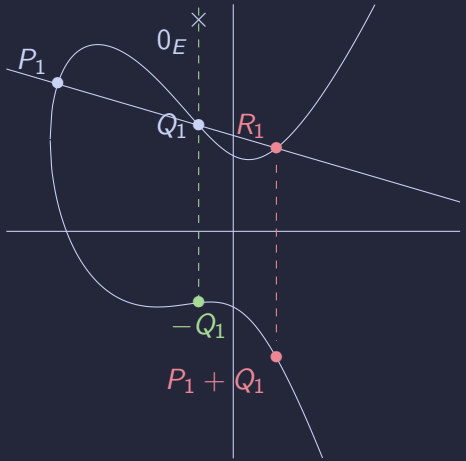

Figure: A binary elliptic curve

Efficient arithmetic

How to compute $P + Q$ and $n \cdot P$ efficiently?
We forget the y-coordinate, more precisely:

$$x : E \to \mathbb{P}^1, \begin{cases} 0_E & \mapsto \infty := (1:0), \\ (x,y) & \mapsto x = \frac{X}{Z} := (X:Z)^1. \end{cases}$$

[1] On $\mathbb{P}^1$, $(X:Z) = (\lambda X : \lambda Z)$ for any $\lambda \in k^*$.

Efficient arithmetic

How to compute $P + Q$ and $n \cdot P$ efficiently?
We forget the y-coordinate, more precisely:

$$x : E \to \mathbb{P}^1, \begin{cases} 0_E & \mapsto \infty := (1:0), \\ (x,y) & \mapsto x = \frac{X}{Z} := (X:Z)^1. \end{cases}$$

Two important operations:

- `xADD`: from $x(P)$, $x(Q)$, $x(P-Q)$, compute $x(P+Q)$;
- `xDBL`: from $x(P)$, compute $x(2 \cdot P)$.

$\rightsquigarrow$ Montgomery ladder[2]: evaluate efficiently $x(P) \mapsto x(n \cdot P)$.

[1] On $\mathbb{P}^1$, $(X:Z) = (\lambda X : \lambda Z)$ for any $\lambda \in k^*$.

[2] P. L. Montgomery. "Speeding the Pollard and elliptic curve methods of factorization". In: *Mathematics of Computation* 48 (1987), pp. 243–264

An example

On the curve $E : x^2 + xy = x(x^2 + ax + b^2)$:

$$\texttt{xADD} : \begin{cases} X_{P+Q} = b^2 Z_{P-Q}(X_P X_Q + b^2 Z_P Z_Q)^2 \\ Z_{P+Q} = X_{P-Q}(bX_P Z_Q + bX_Q Z_P)^2 \end{cases}$$

$$\texttt{xDBL} : \begin{cases} X_{2.P} = (X_P^2 + b^2 Z_P^2)^2 \\ Z_{2.P} = (X_P Z_P)^2 \end{cases}$$

Cost:

- `xADD`: $5M + 2S + 3m_0$;
- `xDBL`: $1M + 3S + 1m_0$.

Isogenies

Definition

An *isogeny* $\varphi : E \to E'$ is:

- a surjective morphism of algebraic curves:

$$\varphi(x, y) = (\varphi_x(x, y), \varphi_y(x, y)),\ \varphi_x, \varphi_y \in k(x, y);$$

- that preserves the group law: $\varphi(P + Q) = \varphi(P) + \varphi(Q)$.

Definition

An *isogeny* $\varphi : E \to E'$ is:

- a surjective morphism of algebraic curves:

$$\varphi(x, y) = (\varphi_x(x, y), \varphi_y(x, y)),\ \varphi_x, \varphi_y \in k(x, y);$$

- that preserves the group law: $\varphi(P + Q) = \varphi(P) + \varphi(Q)$.

Examples:

- Multiplication by $n \in \mathbb{Z}$: $[n] : E \to E, P \mapsto n \cdot P$;
- $\varphi : E : y^2 + xy = x(x^2 + \omega + 1) \to E' : y^2 + xy = x(x^2 + \omega^2 + \omega + 1)$ over $\mathbb{F}_8 = \mathbb{F}_2(\omega)$ given by

$$(x, y) \mapsto \left(\frac{x^3 + (\omega^2 + \omega)x}{(x + \omega)^2}, \cdots \right).$$

Some remarks

$$\varphi : E : y^2 + xy = x(x^2 + \omega + 1) \to E' : y^2 + xy = x(x^2 + \omega^2 + \omega + 1) :$$
$$(x, y) \mapsto \left(\frac{x^3 + (\omega^2 + \omega)x}{(x + \omega)^2}, \cdots \right).$$

1. The x-coordinate of the image only depends on x: $\varphi(x, y) = (\varphi_x(x), \cdots)$.

Some remarks

$$\varphi : E : y^2 + xy = x(x^2 + \omega + 1) \to E' : y^2 + xy = x(x^2 + \omega^2 + \omega + 1) :$$
$$(x, y) \mapsto \left(\frac{x^3 + (\omega^2 + \omega)x}{(x + \omega)^2}, \cdots \right).$$

1. The x-coordinate of the image only depends on x: $\varphi(x, y) = (\varphi_x(x), \cdots)$.
2. If $\varphi_x(x) = \frac{u(x)}{v(x)}$, $\deg(\varphi) := \max(\deg u, \deg v)$.
3. The isogeny is completely determined by its kernel, here $\ker \varphi = \langle(\omega, \omega^2)\rangle$.

Some remarks

$$\varphi : E : y^2 + xy = x(x^2 + \omega + 1) \to E' : y^2 + xy = x(x^2 + \omega^2 + \omega + 1) :$$
$$(x, y) \mapsto \left(\frac{x^3 + (\omega^2 + \omega)x}{(x + \omega)^2}, \cdots \right).$$

1. The x-coordinate of the image only depends on x: $\varphi(x, y) = (\varphi_x(x), \cdots)$.
2. If $\varphi_x(x) = \frac{u(x)}{v(x)}$, $\deg(\varphi) := \max(\deg u, \deg v)$.
3. The isogeny is completely determined by its kernel, here $\ker \varphi = \langle (\omega, \omega^2) \rangle$.
4. φ factors as $\varphi_1 \circ \cdots \circ \varphi_r$ where φ_i is:
   - A scalar multiplication $[n]$: handled with efficient arithmetic.

Some remarks

$$\varphi : E : y^2 + xy = x(x^2 + \omega + 1) \to E' : y^2 + xy = x(x^2 + \omega^2 + \omega + 1) :$$
$$(x, y) \mapsto \left(\frac{x^3 + (\omega^2 + \omega)x}{(x + \omega)^2}, \cdots \right).$$

1. The x-coordinate of the image only depends on x: $\varphi(x, y) = (\varphi_x(x), \cdots)$.
2. If $\varphi_x(x) = \frac{u(x)}{v(x)}$, $\deg(\varphi) := \max(\deg u, \deg v)$.
3. The isogeny is completely determined by its kernel, here $\ker \varphi = \langle (\omega, \omega^2) \rangle$.
4. φ factors as $\varphi_1 \circ \cdots \circ \varphi_r$ where φ_i is:
   - A scalar multiplication $[n]$: handled with efficient arithmetic.
   - Powers of the Frobenius $\pi : (x, y) \mapsto (x^2, y^2)$.

Some remarks

$$\varphi : E : y^2 + xy = x(x^2 + \omega + 1) \to E' : y^2 + xy = x(x^2 + \omega^2 + \omega + 1) :$$
$$(x, y) \mapsto \left(\frac{x^3 + (\omega^2 + \omega)x}{(x + \omega)^2}, \cdots \right).$$

1. The x-coordinate of the image only depends on x: $\varphi(x, y) = (\varphi_x(x), \cdots)$.
2. If $\varphi_x(x) = \frac{u(x)}{v(x)}$, $\deg(\varphi) := \max(\deg u, \deg v)$.
3. The isogeny is completely determined by its kernel, here $\ker \varphi = \langle (\omega, \omega^2) \rangle$.
4. φ factors as $\varphi_1 \circ \cdots \circ \varphi_r$ where φ_i is:
   - A scalar multiplication $[n]$: handled with efficient arithmetic.
   - Powers of the Frobenius $\pi : (x, y) \mapsto (x^2, y^2)$.
   - Separable isogenies of prime degree ℓ.

Some remarks

$$\varphi : E : y^2 + xy = x(x^2 + \omega + 1) \to E' : y^2 + xy = x(x^2 + \omega^2 + \omega + 1) :$$

$$(x, y) \mapsto \left(\frac{x^3 + (\omega^2 + \omega)x}{(x + \omega)^2}, \cdots \right).$$

1. The x-coordinate of the image only depends on x: $\varphi(x, y) = (\varphi_x(x), \cdots)$.
2. If $\varphi_x(x) = \frac{u(x)}{v(x)}$, $\deg(\varphi) := \max(\deg u, \deg v)$.
3. The isogeny is completely determined by its kernel, here $\ker \varphi = \langle(\omega, \omega^2)\rangle$.
4. φ factors as $\varphi_1 \circ \cdots \circ \varphi_r$ where φ_i is:
   - A scalar multiplication $[n]$: handled with efficient arithmetic.
   - Powers of the Frobenius $\pi : (x, y) \mapsto (x^2, y^2)$.
   - Separable isogenies of prime degree ℓ.

Vélu formulas (ℓ odd prime)

- $E : y^2 + xy = x(x^2 + ax + b^2)$;
- $P \in E$ an ℓ-torsion point, $\ell \cdot P = 0_E$;
- $\varphi : E \to E'$ with kernel $\langle P \rangle$, $\varphi(x, y) = (\varphi_x(x), \ldots)$;
- $S = \{x(P), x(2 \cdot P), \ldots, x(\frac{\ell-1}{2} \cdot P)\}$.

Vélu formulas (ℓ odd prime)

- $E : y^2 + xy = x(x^2 + ax + b^2)$;
- $P \in E$ an ℓ-torsion point, $\ell \cdot P = 0_E$;
- $\varphi : E \to E'$ with kernel $\langle P \rangle$, $\varphi(x, y) = (\varphi_x(x), \ldots)$;
- $S = \{x(P), x(2 \cdot P), \ldots, x(\frac{\ell-1}{2} \cdot P)\}$.

Vélu[3] provided formulas to compute E' and φ, in particular

$$E' : y^2 + xy = x(x^2 + ax + b^2 + v + \sqrt{v}), \quad v = \sum_{Q \in S} X_Q / Z_Q,$$

$$\varphi_x(X/Z) = \frac{X}{Z}\left(1 + \sum_{Q \in S} \frac{X_Q/Z_Q}{(X/Z + X_Q/Z_Q)^2}\right).$$

Not adapted for projective formulas!

[3] Jacques Vélu. "Isogénies Entre Courbes Elliptiques". In: *Comptes-Rendus de l'Académie des Sciences* 273 (1971), A238–A241

Projective formulas

- $P \in E$ an ℓ-torsion point, $\ell \cdot P = 0_E$ (ℓ odd prime);
- $\varphi : E \to E'$ with kernel $\langle P \rangle$, $\varphi(x, y) = (\varphi_x(x), \ldots)$;
- $S = \{x(P), x(2 \cdot P), \ldots, x(\frac{\ell-1}{2} \cdot P)\}$.

Theorem (char $k \neq 2$, Hisil–Costello '17, Renes '18)

With notations above, if $T \in E(\overline{k}) \setminus G$, then for some $\alpha \in \overline{k}^*$:

$$\varphi_x(x) - \varphi_x(x_T) = \alpha(x - x_T)\Big[\prod_{Q \in S} \frac{x - x_{Q+T}}{x - x_Q}\Big]^2$$

Projective formulas

- $P \in E$ an ℓ-torsion point, $\ell \cdot P = 0_E$ (ℓ odd prime);
- $\varphi : E \to E'$ with kernel $\langle P \rangle$, $\varphi(x, y) = (\varphi_x(x), \ldots)$;
- $S = \{x(P), x(2 \cdot P), \ldots, x(\frac{\ell-1}{2} \cdot P)\}$.

Theorem (~~char $k \neq 2$~~, Hisil–Costello '17, Renes '18)

With notations above, if $T \in E(\overline{k}) \setminus G$, then for some $\alpha \in \overline{k}^*$:

$$\varphi_x(x) - \varphi_x(x_T) = \alpha(x - x_T)\Big[\prod_{Q \in S} \frac{x - x_{Q+T}}{x - x_Q}\Big]^2$$

Generalizes to char $k = 2$ (Banegas, S., Smith '26).

An example

$$E : y^2 + xy = x(x^2 + ax + b^2) \text{ and } E' : y^2 + xy = x(x^2 + ax + b'^2)$$

- we choose $T = (0, 0) \notin G$ as the special point;
- we split numerator and denominator $\varphi_x(X/Z) = (\varphi_X(X : Z) : \varphi_Z(X : Z))$;
- b' is computed with Vélu formulas;

$$\varphi_X(X : Z) = b'X\Big[\prod_{Q \in S} (XX_Q + (bZ)(bZ_Q))\Big]^2$$

$$\varphi_Z(X : Z) = bZ\Big[\prod_{Q \in S} (X(bZ_Q) + X_Q(bZ))\Big]^2.$$

Benchmarks

Table: Average cycles for one x-evaluation in $\mathbb{F}_{2^{511}}$, averaged over 50×32 calls per ℓ.

ℓ	Vélu	Projective	With normalisation
3	10 393 169	75 216	10 543 881
5	19 621 248	125 393	10 555 526
7	32 175 389	160 394	10 010 506
13	60 861 058	387 269	10 945 638
17	79 744 703	433 497	10 716 132
19	93 595 902	418 651	10 220 777
29	139 227 239	639 577	10 439 875

Average speed-up of 24.41× (or 4.22× with normalisation).

A note on supersingular curves

- Ordinary curves: E such that there is $T \neq 0_E$ verifying $T = -T$:

$$E : y^2 + xy = x(x^2 + ax + b^2),\ b \neq 0 \qquad T = (0, 0)$$

A note on supersingular curves

- Ordinary curves: E such that there is $T \neq 0_E$ verifying $T = -T$:

$$E : y^2 + xy = x(x^2 + ax + b^2),\ b \neq 0 \qquad T = (0,0)$$

- On $\overline{k}$, there is only one supersingular curve in char. 2! (up to isomorphism)

$$E : y^2 + y = x^3$$

$\rightsquigarrow$ every isogeny is an endomorphism, and we know the endomorphism ring ✓

Conclusion

Conclusion

Summary

- Exhaustive computation of isogenies in char 2.
- Generalized the projective formula in char 2.
- Comparison of performances in C.

Perspective

Including this work to a CRS implementation in characteristic 2.